Summary

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A short summary on how to formalize:

Theorem 1 (Edmonds Gallai)**.** *Every graph $G = (V, E)$ contains a vertex set S with the following two properties:*

i) S is matchable to the components of $G[S^c]$

ii) Every component of $G[S^c]$ is factor-critical

Given any such set S, the graph G contains a 1-factor if and only if $|S| = |\mathcal{C}_{G-S}|$.

We will be working with a type `V` which is an instance of `Fintype V`, to model graphs we use `G : SimpleGraph V`. Connected components of a `SimpleGraph` are expressed as `ConnectedComponent`. Given a `A : Set V` the induced graph `G.induce A` is a `SimpleGraph` over the type `{ x // x ∈ A }` which inherits the adjacency relation of the ambient graph `G`.
To fully state the theorem it remains to define what it means for a `S : Set V` to be *matchable* to components of `G.induce S`^{c}. For that we will deviate from the definition in the book [1] and instead use the following:

```
def IsMatchableToComponents (G : SimpleGraph V) (S : Set V) : Prop :=
  ∃ (f : S → (G.induce Sᶜ).ConnectedComponent),
  Function.Injective f ∧ (∀ s : S, ∃ y ∈ (f s), G.Adj ↑s ↑y)
```

One advantage this definition has is the ability to make use of the already formalized marriage theorem if `Fintype V` holds. Similarly we refrain from following the graph-centric definition in the book of being *factor-critical* in favor of using sets:

```
def IsFactorCriticalArea (G : SimpleGraph V) (S : Set V) : Prop :=
  S.Nonempty ∧
  ∀ v ∈ S, ∃ M : G.Subgraph, M.IsMatching ∧ M.support = S \ {v}
```

The overall attack-plan will be to (almost) mimic the proof of the book. As in the book we will consider sets $T \subset V$ where Tutte's condition fails the worst, i.e sets where:

$$d(T) := q(G[T^c]) - |T|$$

is maximal.

We define `d` is analogously:

```
def d (G : SimpleGraph V) (S : Set V) : ℤ :=
    (G.induce Sᶜ).oddComponents.ncard - S.ncard
```

A set S is said to be *Edmonds-Gallai*, if it is the largest such T:

```
def IsEdmondsGallai (G : SimpleGraph V) (S : Set V) : Prop :=
  ∀ B : Set V, (d G S ≥ d G B) ∧ (d G S = d G B → S.ncard ≥ B.ncard)
```

Because tutte's theorem is already formalized in lean, we can avoid the proof by induction in the book. Given the existence of an `S : Set V` which is Edmonds-Gallai, we first prove that (the support of) every component of `G.induce Sᶜ` is a factor-critical area. Then, by the marriage theorem if `S` is not matchable to the components of `G.induce Sᶜ` there exists a `T : Set S` that sends fewer than `T.ncard` edges to the components of `G.induce Sᶜ`. Under the assumption that `T` exists, we prove that `G.d (S \ T) > G.d S` contradicting the fact that `S` is Edmonds-Gallai:

```
theorem aux (G : SimpleGraph V) : ∃ (S : Set V),
  (G.IsMatchableToComponents S) ∧
  (∀ (C : (G.induce Sᶜ).ConnectedComponent),
  (G.induce Sᶜ).IsFactorCriticalArea C.supp) := by
  classical
  rcases G.exists_isEdmondsGallai with ⟨S, hS⟩
  refine ⟨S, ⟨?_, hS.isFactorCriticalArea_supp⟩⟩
  by_contra! h
  rcases not_matchable_exists_hall_violator h with ⟨T, hT⟩
  linarith[G.deficiency_remove_hall_violator_lt T hT, hS (S \ T)]
```

Finally, the iff. statement of our theorem is stated as:

```
lemma exists_isPerfectMatching_iff_card_eq
  (h₀ : G.IsMatchableToComponents S)
  (h₁ : ∀ (C : (G.induce Sᶜ).ConnectedComponent),
  (G.induce Sᶜ).IsFactorCriticalArea C.supp) :
  card S = card (G.induce Sᶜ).ConnectedComponent ↔
  ∃ M : Subgraph G, M.IsPerfectMatching
```

If `card S = card (G.induce Sᶜ).ConnectedComponent`, the function `f` yielded by `G.IsMatchableToComponents S` is bijective and one constructs a perfect matching in the obvious way. Conversely if `G` admits a perfect matching, it follows by tutte that `card S ≥ card (G.induce Sᶜ).ConnectedComponents`, since `S` is matchable to the components of `G.induce Sᶜ` we have `card S ≤ card (G.induce Sᶜ).ConnectedComponents` and hence equality.

In the following we will expand upon the main lemmas of our proof that is `isFactorCriticalArea_supp` and `deficiency_remove_hall_violator`. As each of these follow from a short calculation, we will provide the main lemmas / theorems in lean that justify the steps.

1 Equations on Components

The proof in the book, relies on the fact that given a graph $G = (V, E)$, a component C of G and $T \subset C$, one can express the number of odd components of $G[T^c]$ as:

$$q(G[T^c]) = q(G[C^c]) + q(G[C \setminus T])$$

To formalize this statement, we say that a set $S \subset V$ is *closed* if there exists no edge $xy \in E$, such that $x \in S$ and $y \in S^c$.

```
def IsClosed (G : SimpleGraph V) (S : Set V) : Prop :=
    ¬∃ x ∈ S, ∃y ∈ Sᶜ, G.Adj x y
```

Note that any component C of G must be a subset of S or S^c. If not, there would exist a $S - S^c$-walk in C and thus an $S - S^c$-edge. Conversely if C is a component of $G[S]$, then C must also be a component of G. As C is connected, there exists a component $\tilde{C}$ of G, such that $C \subset \tilde{C}$. Since $\tilde{C}$ lies in S it is also a component of $G[S]$, by maximality it follows that $C = \tilde{C}$. For any closed set S we thus have:

$$q(G) = q(G[S]) + q(G[S^c])$$

The theorem `oddComponents_ncard_add_compl_eq` states this fact in lean:

```
theorem oddComponents_ncard_add_compl_eq [Fintype V]
(h : G.IsClosed S) :
  (G.induce S).oddComponents.ncard + (G.induce Sᶜ).oddComponents.ncard =
    G.oddComponents.ncard := ...
```

the above proof for the theorem, also follows in lean - given a `S : Set V` which is closed in `G : SimpleGraph V` the embedding `G.induce S ↪g G` induces an injective function `(G.induce S).ConnectedComponent → G.ConnectedComponent` which preserves the underlying support and order of the components.

More generally, let T be a set such that $T \subset S$, since S^c is closed and $S^c \subset T^c$ we have:

$$\begin{aligned} q(G[T^c]) &= q(G[T^c][S^c]) + q(G[T^c][T^c \cap S]) \\ &= q(G[S^c]) + q(G[S \setminus T]) \end{aligned}$$

The lemma `oddComponents_ncard_eq` proves the fact that isomorphic graphs have the same amount of odd components:

```
lemma Iso.oddComponents_ncard_eq (φ : G ≃g G') :
     G.oddComponents.ncard = G'.oddComponents.ncard := ...
```

Using appropriate isomorphisms, we can obtain `ncard_oddComponents_induce_compl_eq_add'`, by the very same proof:

```
lemma ncard_oddComponents_induce_compl_eq_add'
  (h : S ⊆ T) (hT : G.IsClosed T) :
  (G.induce Sᶜ).oddComponents.ncard = (G.induce Tᶜ).oddComponents.ncard +
     (G.induce (T \ S)).oddComponents.ncard := ...
```

2 $C \in \mathcal{C}$ is a factor-critical area

The proof of the components being factor-critical areas, depends as in the book, on the fact that given an `S : Set V` which is Edmonds-Gallai, the components of `G.induce S`^{c} are of odd order. Our proof odd_ncard_supp, will follow the book to the tee.

```
lemma IsEdmondsGallai.odd_ncard_supp (h : G.IsEdmondsGallai S) :
  ∀ C : (G.induce Sᶜ).ConnectedComponent, Odd C.supp.ncard := ...
```

Suppose that there exists a component C of $G[S^c]$ which has an even order, pick a $c \in C$ and let $T := S \cup \{c\}$, we show that $d(T) \geq d(S)$ contradicting the choice of S. The fact then follows from the calculation:

$$
\begin{aligned}
d(T) &= q(G[T^c]) - |T| \\
&= q(G[S^c][S^c \setminus \{c\}]) - |S| - 1 \\
&\quad \texttt{by Iso.oddComponents_ncard_eq} \\
&= q(G[S^c][C^c]) + q(G[S^c][C \setminus \{c\}]) - |S| - 1 \\
&\quad \texttt{by ncard_oddComponents_induce_compl_eq_add} \\
&\geq q(G[S^c][C^c]) + 1 - |S| - 1 \\
&\quad \texttt{by odd_ncard_geq_one_oddComponents} \\
&= q(G[S^c][C^c]) + q(G[S^c][C]) - |S| \\
&\quad \texttt{by odd_comp_eq_zero_induce_even_comp} \\
&= q(G[S^c]) - |S| \\
&\quad \texttt{by oddComponents_ncard_add_compl_eq} \\
&= d(S)
\end{aligned}
$$

where odd_ncard_geq_one_oddComponents is a proof that graphs of odd order have at least one odd component and odd_comp_eq_zero_induce_even_comp states that $q(G[C]) = 0$ for any component of even order.

Using the above lemma, we are able to prove that the support of every component of $G[S^c]$ is a factor-critical area. The full statement of `isFactorCriticalArea_supp` is:

```
lemma IsEdmondsGallai.isFactorCriticalArea_supp
  (h : G.IsEdmondsGallai S) :
  ∀ C : (G.induce Sᶜ).ConnectedComponent,
  (G.induce Sᶜ).IsFactorCriticalArea C.supp := ...
```

Again suppose for a contradiction, that there exists a component C of $G[S^c]$ which is not a factor-critical area. Then there exists a $c \in C$, such that $G[C \setminus \{c\}]$ has no perfect matching. By tutte's theorem, there exists a set $Q \subset C \setminus \{c\}$ such that $q(G[C \setminus \{c\}][C \setminus (Q \cup \{c\}]) > |Q|$. Setting $T := S \cup Q \cup \{c\}$ we again show that $d(T) \geq d(S)$.

$$
\begin{aligned}
d(T) &= q(G[T^c]) - |T| \\
&= q(G[S^c][S^c \setminus (Q \cup \{c\})]) - |S| - |Q| - 1 \\
&\quad \texttt{by Iso.oddComponents_ncard_eq} \\
&= q(G[S^c][C^c]) + q(G[S^c][C \setminus (Q \cup \{c\})]) - |S| - |Q| - 1 \\
&\quad \texttt{by ncard_oddComponents_induce_compl_eq_add} \\
&= q(G[S^c][C^c]) + q(G[C \setminus \{c\}][C \setminus (Q \cup \{c\}]) - |S| - |Q| - 1 \\
&\quad \texttt{by Iso.oddComponents_ncard_eq} \\
&\geq q(G[S^c][C^c]) + 1 - |S| \\
&\quad \texttt{by oddComponents_sub_ncard_le_two} \\
&= q(G[S^c][C^c]) + q(G[S^c][C]) - |S| \\
&\quad \texttt{by odd_comp_eq_one_induce_odd_comp} \\
&= q(G[S^c]) - |S| \\
&\quad \texttt{by oddComponents_ncard_add_compl_eq} \\
&= d(S)
\end{aligned}
$$

The lemma `oddComponents_sub_ncard_le_two` proves that given a graph G of even order and a set S such that $q(G[S^c]) - |S| > 0$, then $q(G[S^c]) - |S| \geq 2$. Analogously, `odd_comp_eq_one_induce_odd_comp` proves that $q(G[C]) = 1$ for any component of odd order.

3 S is matchable to the components of $G[S^c]$

Given a `S : Set V` that is not matchable to the components of `G.induce S`^{c}, via the marriage theorem, the lemma `not_matchable_exists_hall_violator` provides the existence of a `T : Set S` that sends fewer than `T.ncard` edges to the components of `G.induce S`^{c}:

```
lemma not_matchable_exists_hall_violator [Fintype V]
  (h : ¬ G.IsMatchableToComponents S) :
  ∃ (A : Set S),
     A.ncard > (⋃ a ∈ A, G.connectedComponentsNeighbors a).ncard := ...
```

where `G.connectedComponentsNeighbors a` is the set of connected components that share an edge with `a`:

```
def connectedComponentsNeighbors (s : S) :
   Set (G.induce Sᶜ).ConnectedComponent :=
  {C : (G.induce Sᶜ).ConnectedComponent | ∃ y ∈ C.supp, G.Adj s y}
```

Finally, let T be such a set, then our lemma `deficiency_remove_hall_violator_lt` proves the desired result:

```
lemma deficiency_remove_hall_violator_lt
  (T : Set S)
  (hT : T.ncard > (⋃ t ∈ T, G.connectedComponentsNeighbors t).ncard) :
  d G S < d G (S \ ↑T) := ...
```

Let $\mathcal{C}_T$ be the union of all components in $G[S^c]$ that share an edge with T. By assumption on T, we have that $|T| > q(G[S^c][\mathcal{C}_T])$. Since $S^c \setminus \mathcal{C}_T$ is closed in $G[S^c]$ and shares no edge with T, it must also be closed in $G[(S \setminus T)^c]$. Then by calculation we see that:

$$
\begin{aligned}
d(S \setminus T) &= q(G[(S \setminus T)^c]) - |S \setminus T| \\
&\geq q(G[(S \setminus T)^c][S^c \setminus \mathcal{C}_T]) - |S| + |T| \\
&\qquad \texttt{by oddComponents_ncard_add_compl_eq} \\
&= q(G[S^c][S^c \setminus \mathcal{C}_T]) - |S| + |T| \\
&\qquad \texttt{by Iso.odd_comp_card_eq} \\
&= q(G[S^c]) - q(G[S^c][\mathcal{C}_T]) - |S| + |T| \\
&\qquad \texttt{by oddComponents_ncard_add_compl_eq} \\
&> q(G[S^c]) - |S| \\
&= d(S)
\end{aligned}
$$

References

[1] Reinhard Diestel. *Graph Theory, 5th Edition*. Springer, 2017.